

Architecture

a Biomni-inspired tool registry driven by an AI-style agent loop — grounded, reasoner-verified, curator-in-the-loop

input cell-type name · markers · expression matrix · location

agent loop **retrieve → plan → execute → self-correct → critique**

self-correct loops back to execute on failure; plan + define call the optional LLM

Tool registry

ols_search · literature_search · marker_panel (NS-Forest) · go_marker_support · draft_definition · taxon_constraints — declarative ToolSpecs

Reasoning

ROBOT · ELK — classifies the draft against cl-base.owl (the whole Cell Ontology) and flags duplicates of existing terms

CL-native outputs

KGCL · MIRACL · SSSOM · ROBOT template / OWL · GitHub new-term issue

Grounding & evidence

EBI OLS4 · Europe PMC · QuickGO · NCBITaxons — every identifier resolved to a real ontology term

LLM — optional

Groq (free) · OpenAI · Anthropic — plans the tool order and phrases the definition; never invents ontology terms

Ecosystem hand-offs

OntoGPT / SPIRES · CurateGPT / DRAGON-AI · Aurelian — detect, defer, fall back